

CAID SoC — What We Built & How the Project Is Organized

The Project

- A complete SoC built around a custom RISC-V (RV32I) core
- Core + memory + GPIO connected over a Wishbone bus
- GCC-compatible — runs real, compiled C programs
- Taken from RTL design through simulation, real FPGA hardware, and a full ASIC layout

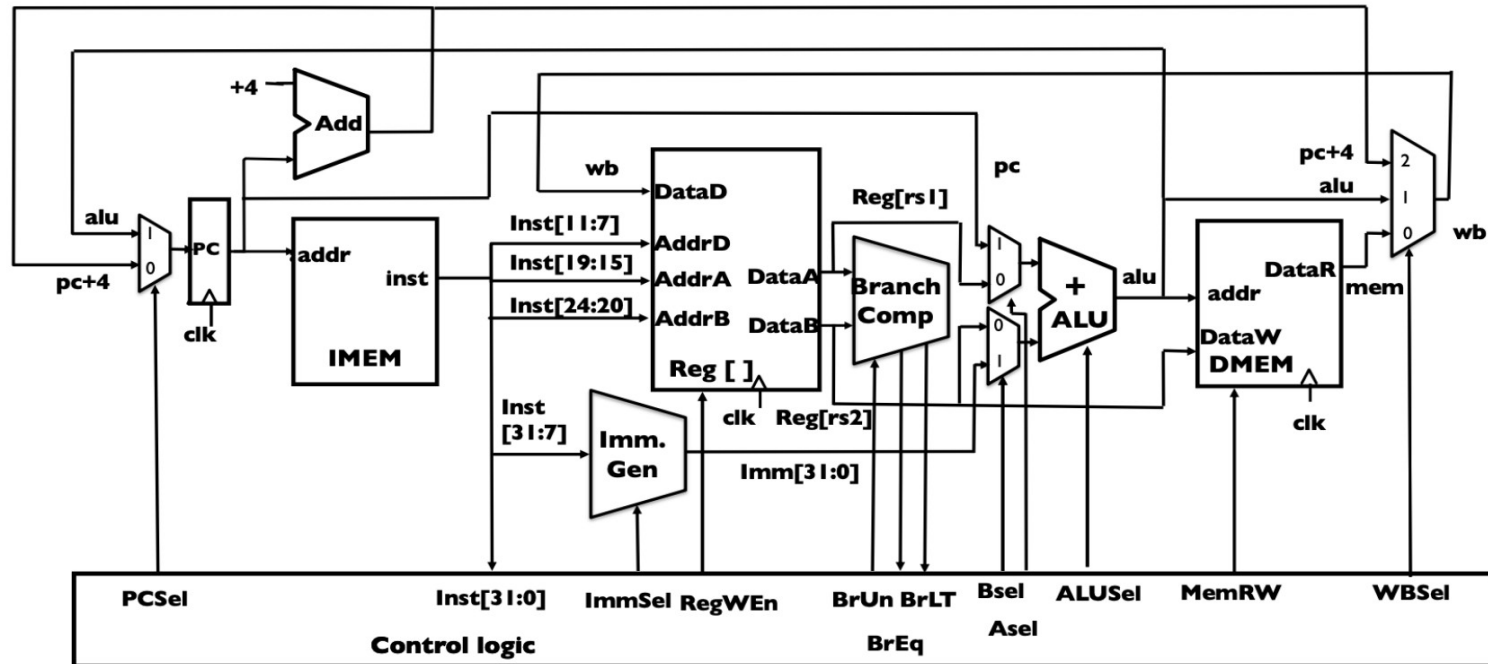
This Project Is Divided six phases



Core Design — Single-Cycle RV32I RISC-V Core

What We Designed

- A single-cycle RISC-V core, designed from scratch in Verilog
- Full Datapath: PC, ALU, Register File, Immediate Generator, Branch Comparator, Control Unit

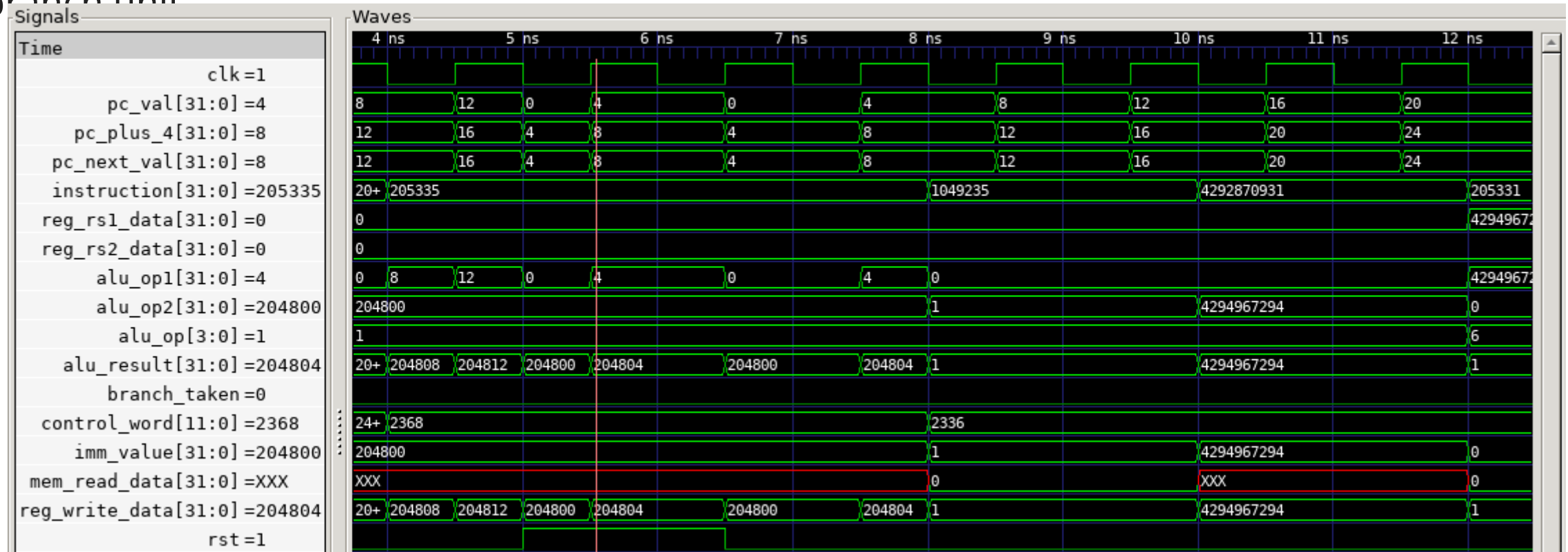


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Core Verification & Validation

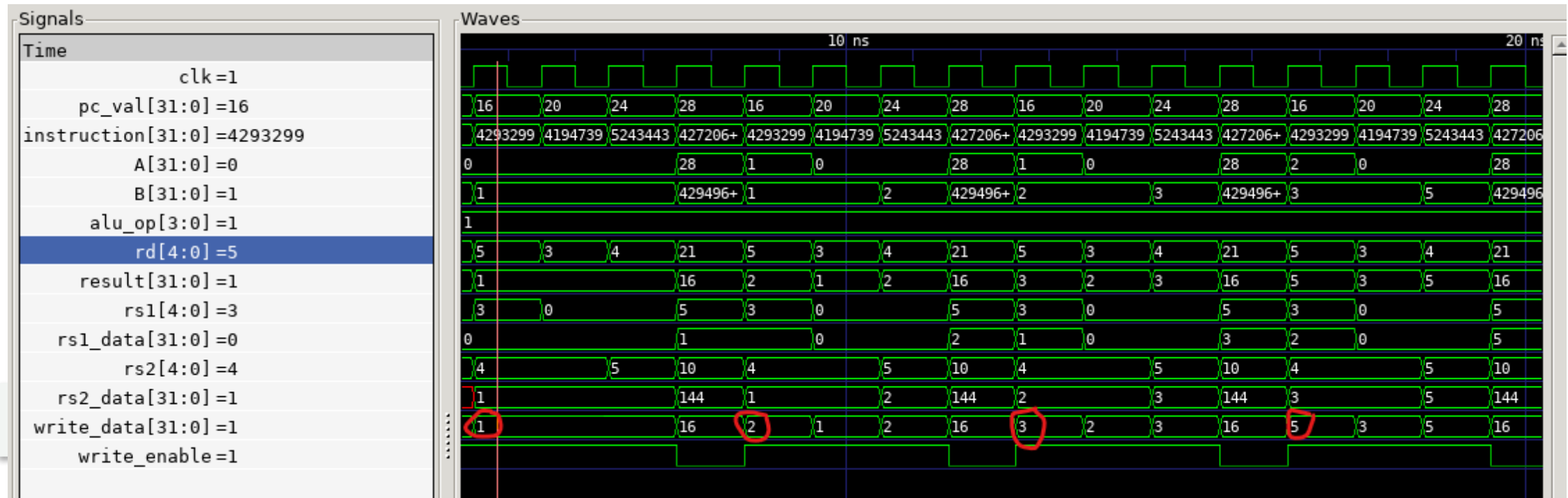
Full RV32I instruction set — R, I, S, B, U & J-type

Functional units individually: ALU, register file, immediate generator, branch unit

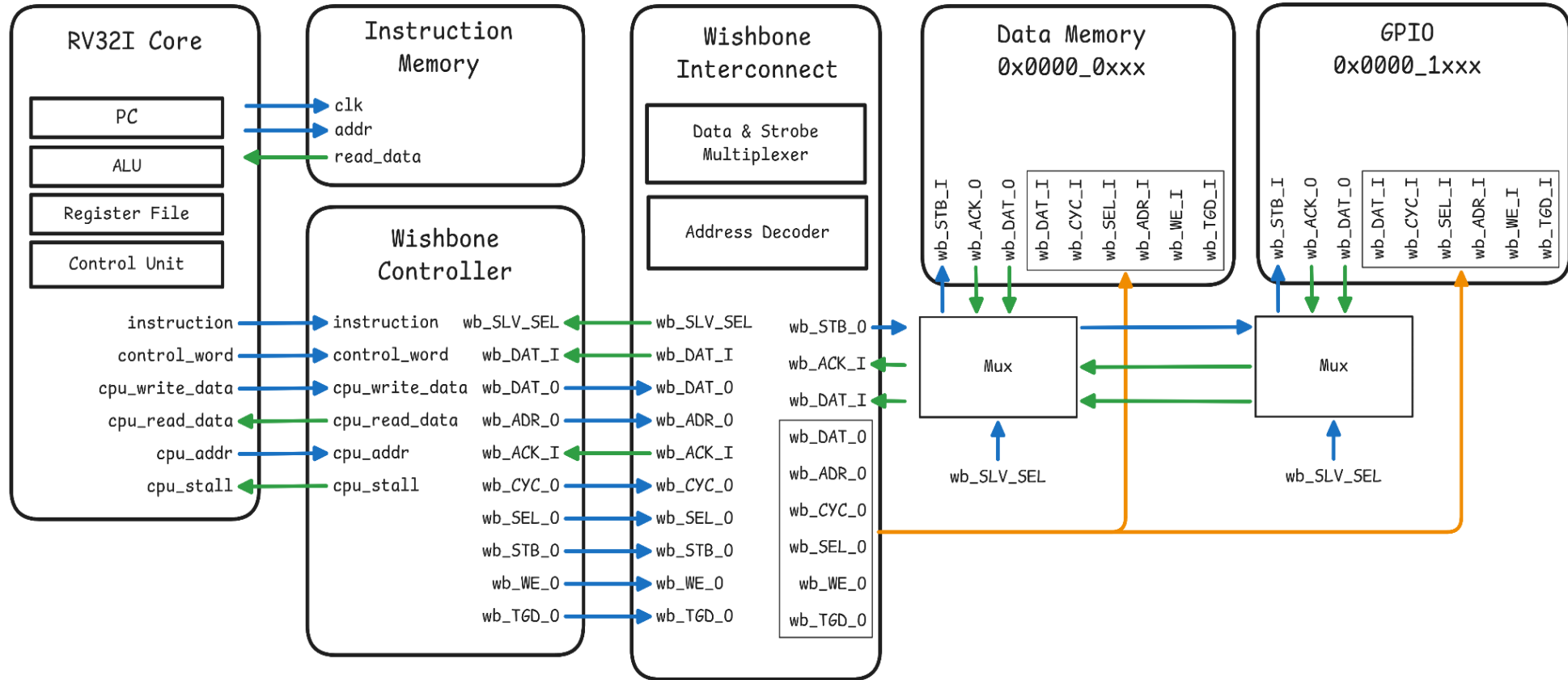


Software Ecosystem Enablement — GCC Compatibility

C Source → RISC-V GCC → ELF Binary → .mem Hex File → Instruction Memory → RISC-V Core Execution → GTKWave Verification → Fibonacci Result (1, 2, 3, 5, 8 ... 144)



SoC & Wishbone Integration



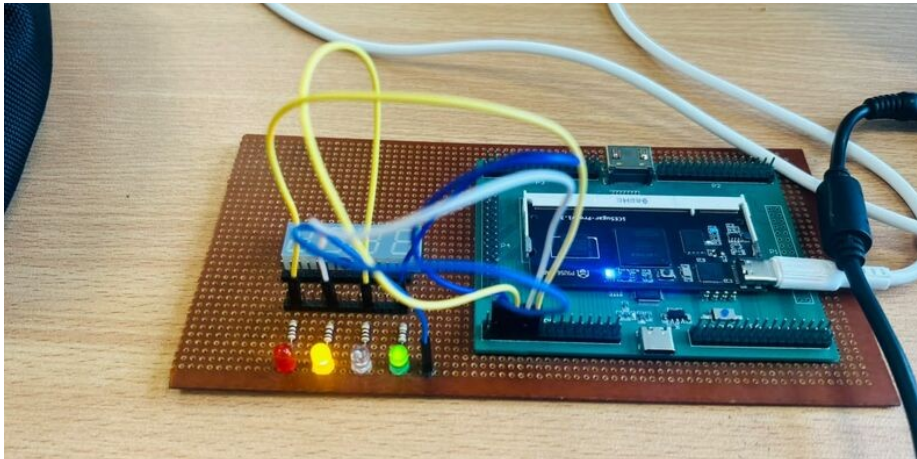
Multi-FPGA Deployment

RTL → Synthesis → Place & Route → Bitstream → Hardware Testing

- **Synthesis:** Yosys
- **Place & Route:** nextpnr
- **Programming:** openFPGALoader

ECP5 Colorlight i9

- **FPGA:** Lattice ECP5 LFE5U-45F (CABGA381)
- **Onboard Clock:** 25 MHz
- **Timing-Closure Target:** 65 MHz
- **LUT Utilization:** 1193 / 43848 (2%)



```
Info: Device utilisation:
Info:      TRELIS_IO:      8/    245    3%
Info:      DCCA:           1/     56    1%
Info:      DP16KD:          0/    108    0%
Info:      MULT18X18D:      0/     72    0%
Info:      ALU54B:          0/     36    0%
Info:      EHXPPLL:         0/      4    0%
Info:      EXTREFB:          0/      2    0%
Info:      DCUA:            0/      2    0%
Info:      PCSCLKDIV:        0/      2    0%
Info:      IOLOGIC:          0/    160    0%
Info:      SIOLOGIC:         0/     85    0%
Info:      GSR:             0/      1    0%
Info:      JTAGG:            0/      1    0%
Info:      OSCG:            0/      1    0%
Info:      SEDGA:           0/      1    0%
Info:      DTR:             0/      1    0%
Info:      USRMCLK:          0/      1    0%
Info:      CLKDIVF:          0/      4    0%
Info:      ECLKSYNCB:        0/     10    0%
Info:      DLLDELD:          0/      8    0%
Info:      DDRDLL:           0/      4    0%
Info:      DQSBUFM:          0/     10    0%
Info:      TRELIS_ECLKBUF:    0/      8    0%
Info:      ECLKBRIDGECS:     0/      2    0%
Info:      DCSC:            0/      2    0%
Info:      TRELIS_FF:       44/  43848    0%
Info:      TRELIS_COMB:    1193/  43848    2%
Info:      TRELIS_RAMW:     32/   5481    0%
```

RISC-V Core
Development

Core Verification &
Validation

Software Ecosystem
Enablement

SoC & Wishbone
Integration

Multi-FPGA
Deployment

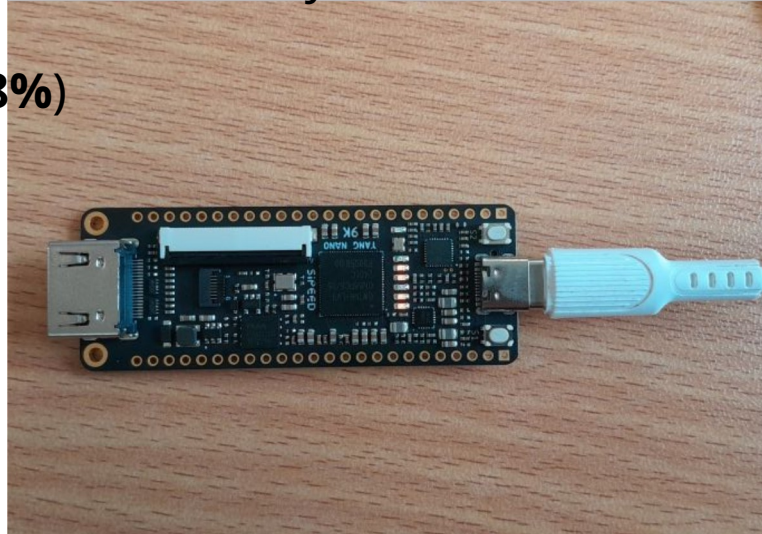
Multi-FPGA Deployment

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- **Synthesis:** Yosys
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- **Programming:** openFPGALoader

Sipeed Tang Nano 9K

- **FPGA:** Gowin GW1N-9C
- **Onboard Clock:** 27 MHz
- **PLL:** Not used; oscillator drives the SoC directly
- **Post-P&R Fmax:** 32.57 MHz
- **LUT Utilization:** 1611 / 8,640 (18%)



```
Info: Device utilisation:
Info:          VCC:      1/      1 100%
Info:          IOB:      7/    274   2%
Info:          LUT4:    1611/   8640  18%
Info:          OSER16:    0/     80   0%
Info:          IDES16:    0/     80   0%
Info:          IOLOGICI:  0/    276   0%
Info:          IOLOGICO:  0/    276   0%
Info:          MUX2_LUT5:  590/   4320  13%
Info:          MUX2_LUT6:  280/   2160  12%
Info:          MUX2_LUT7:  123/   1080  11%
Info:          MUX2_LUT8:   52/   1080   4%
Info:          ALU:      138/   6480   2%
Info:          GND:       1/     1 100%
Info:          DFF:      43/   6480   0%
Info:          RAM16SDP4:  32/    270  11%
Info:          BSRAM:     0/     26   0%
Info:          ALU54D:     0/     10   0%
Info:          MULTADDALU18X18: 0/     10   0%
Info:          MULTALU18X18:  0/     10   0%
Info:          MULTALU36X18:  0/     10   0%
Info:          MULT36X36:     0/     5   0%
Info:          MULT18X18:     0/     20   0%
Info:          MULT9X9:      0/     40   0%
Info:          PADD18:       0/     20   0%
Info:          PADD9:        0/     40   0%
Info:          GSR:         1/     1 100%
Info:          OSC:         0/     1   0%
Info:          rPLL:         0/     2   0%
Info:          FLASH608K:     0/     1   0%
Info:          BUFG:         0/     22   0%
Info:          DQCE:         0/     24   0%
Info:          DCS:          0/     8   0%
Info:          DHCEN:        0/     24   0%
Info:          CLKDIV:        0/     8   0%
Info:          CLKDIV2:       0/     16   0%
```

RISC-V Core
Development

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ASIC & Silicon Readiness — GDS Flow

Design Flow

- LibreLane — open-source RTL-to-GDSII flow

PDK & Library

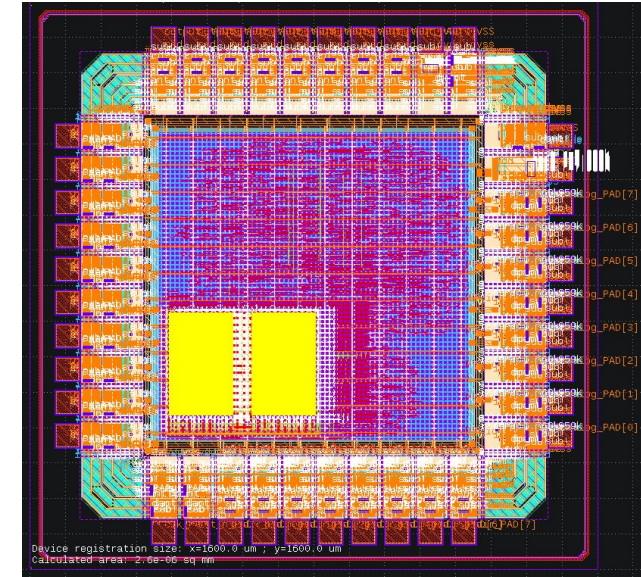
- IHP SG13G2 — 130nm open-source PDK
- sg13g2 standard-cell library + SRAM macro (data memory)

Key Specifications

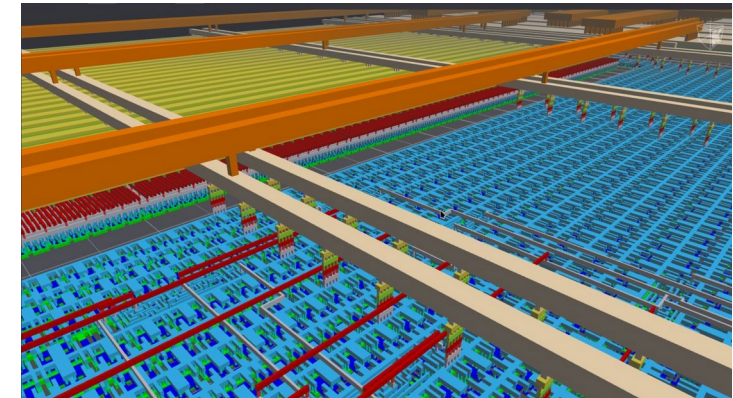
- Clock Target: 50 MHz (86MHz @typ corner)
- Die area: 1.6 mm x 1.6 mm
- ~3969 standard cells placed & routed

Issues

- Slew & Cap Violations
- ~1600 DRC errors
- No programming support



Full-chip layout — 1.6 x 1.6 mm die



3D routing detail view

RISC-V Core
Development

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Software Ecosystem
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SoC & Wishbone
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Multi-FPGA
Deployment

ASIC & Silicon
Readiness

POST-LAYOUT ASIC CORE

RISC-V SoC • RTL-to-GDSII



TECHNOLOGY

IHP SG13G2

130 nm CMOS



LOGIC + MEMORY

3,881 standard cells

1 × SRAM macro

1024 × 32 bits

SRAM counted separately

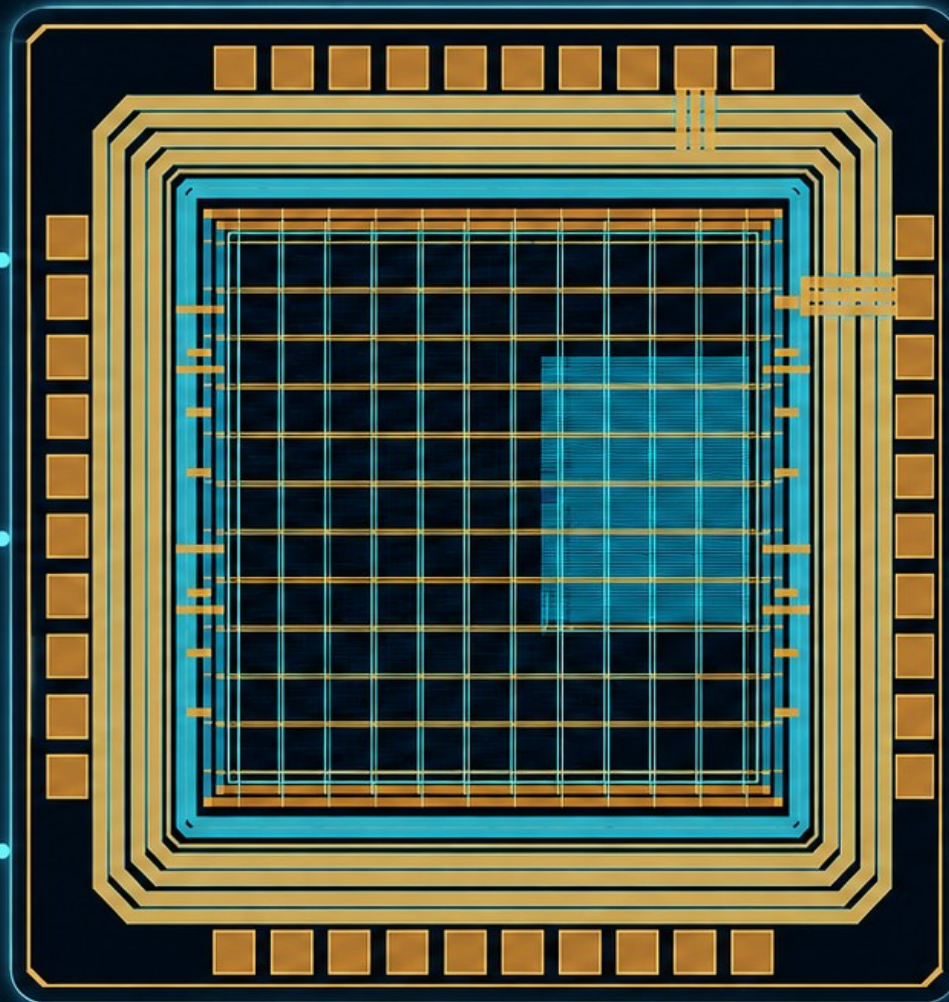


POWER

3.43 mW

Estimated post-layout

Switching: 3.32 mW



DIE + CORE AREA

Die: 1.60 × 1.60 mm

Core: 0.752 mm²



CORE UTILIZATION

26.32%

Std-cell region: 9.45%



CLOCK / TIMING

Closed at 3 PVT corners

Setup margin: +1.11 ns

Hold margin: +0.125 ns

Clock rate not reported



0

SETUP/HOLD
VIOLATIONS



0

ROUTE DRC



0

ANTENNA
VIOLATIONS



WORST IR DROP

0.653 mV



Remaining checks: 13 slew • 32 fanout • 2 density

From Concept to Hardware — The Complete Journey

- Started with a from-scratch, single-cycle RV32I RISC-V core
- Verified against the full ISA, then made GCC-compatible
- Integrated into a complete SoC over a Wishbone bus
- Deployed and proven running on real FPGA hardware
- Carried through a full open-source ASIC flow to a signed-off GDSII layout



Thank You

Questions & Live Demonstration